

High Power Density 1 MHz 3 kW 400 V-48 V LLC Converter for Datacenters with improved Core Loss and Termination Loss

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Abstract— Power consumption of data centers is increasing rapidly over recent years. Recently, 48V power architecture has attracted more interest in data centers as it offers more efficient architecture. This paper focuses on the implementation of the DC-DC stage of the 3 kW power supply unit. Full-bridge regulated LLC converter is designed with a matrix of two transformers. This paper focuses on the magnetic design of the PCB-based transformer and resonant inductor of the LLC converter. The dimensional effects on core loss are discussed and evaluated. A matrix of two transformers implemented on UI core with rectangular core shape is proposed to minimize the eddy core loss of the transformer core. The termination of the secondary Full-bridge rectifier is also optimized to reduce the conduction loss of the transformer. A low-profile prototype of the designed converter is demonstrated with a peak efficiency of 98% and power density of 700W/in³.

Keywords—LLC resonant converter, 48 V bus architecture, Datacenters, Core loss, Dimensional effects.

I. INTRODUCTION

In today's industrial power converters, converter reliability is essential, and converter topologies are well-established. Without a doubt, the power electronics industry continues to seek efficient power delivery and high power-density. Research efforts should be directed towards improving the design practice for the existing topologies to improve fabrication, efficiency,

and reliability. Resonant converters such as LLC and CLLC have been intensively studied and applied in many applications such as data centers and telecom power supplies, EV chargers, TV panels, LED drivers, etc. According to the Report to Congress on Server and Data Center Energy Efficiency [1], the estimate of energy consumption of only data center applications has exceeded 1.5 % of the total U.S. electricity consumption or about \$4.5 billion in electricity costs. Therefore, saving a fraction of that power is substantial.

Datacenters have been switching focus from the 12 V to the 48V power architecture, as shown in Fig. 1, for several reasons [2]. First, the 48V power architecture has less dissipation loss on the bus as it uses higher voltage. Second, a 48V battery bank is connected directly to the 48V bus to support the loads during energy shortage which saves the loss of the extra UPS in the 12 V architecture. Recently, the demand for higher power per server rack is increasing [3] and power supply unit with high power and high-power density is needed. The power supply unit is required to deliver 3 kW or higher with a high power density and high efficiency. A couple of 3 kW power supply products are already introduced with high efficiency up to 98% and power density lower than 50 W/in³ [3] [4].

Resonant converters present an excellent choice for efficient DC-DC converters as they offer soft-switching for both primary and secondary devices. The LLC Resonant converter, along with wide-bandgap WBG devices, breaks new grounds to push the switching frequency to much higher frequencies than those of the traditional pulse width modulation (PWM) converters. The switching frequency in the LLC circuit can now be pushed to 1 MHz or more using GaN devices [5] which opens opportunities to reduce the converter size and push for higher power density. When the magnetic size is reduced, planar

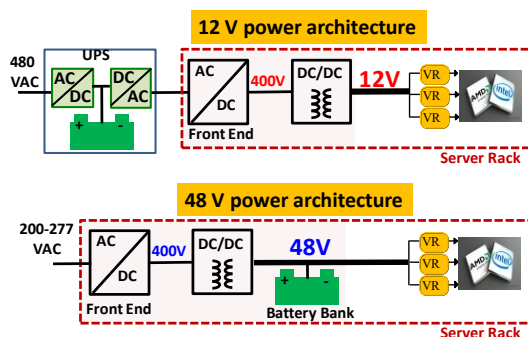


Fig. 1. 12V power architecture and 48V architecture for datacenters

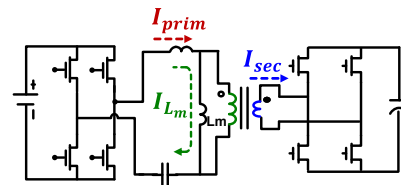


Fig. 2. Full-bridge LLC resonant converter

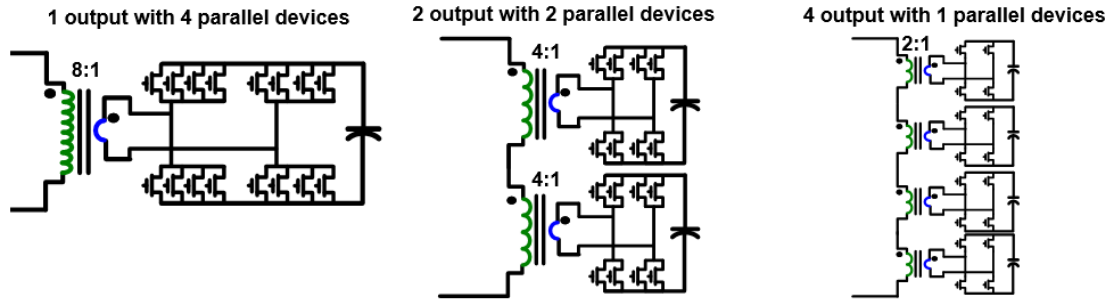


Fig. 3. Three possible matrix transformer designs with different number of output

magnetics (i.e., PCB winding) become more relevant. PCB magnetics feature multiple advantages over Litz wire:

- Fully automated fabrication reduces the cost of the converter.
- PCB windings feature less leakage between transformer windings because of the flexibility of the winding interleaving and the reduced number of turns.
- There is better thermal management due to the increased surface-to-body ratio.
- The design has a low profile and high-power density.

The highly efficient power supply is realized by using a three-phase interleaved LLC converter to reduce conduction loss at such high power. Although the three-phase LLC converter is interesting from the efficiency point of view, still the magnetic design is more complex than the single-phase LLC and requires an extra active current sharing control circuit [6]. Another design based on a single-phase LLC converter is presented in [7] that uses planar magnetic with 6-layer flexible PCB for the primary winding and 12-Layer PCB for the secondary, and Litz-wire resonant inductor. This makes high-efficiency solutions more expensive, and the return back is questioned.

This paper focuses on the design of a high-efficiency high power density 1MHz 3kW full-bridge LLC converter for 48V front-end power supply as shown in Fig. 2. The converter is designed on 6-Layer PCB that includes both primary, secondary, shielding, and inductor winding. The transformer of the 48V resonant converter is excited with four times higher volt.sec compared the 12V converter. This makes the core cross-section area (A_e) is almost 4x higher as well. With increased core size at high frequency, the transformer core exhibits dimensional effects, namely dimensional resonance and skin effect [8]. Moreover, the core eddy loss component tends to increase with increased core size. In fact, [9] shows a linear relationship between core A_e and core's eddy loss. The impact of dimensional effects on core loss is evaluated for this application and quantified during the optimization process of the transformer.

II. DESIGN CONSIDERATIONS FOR HIGH FREQUENCY POWER TRANSFORMER

Single-phase LLC converter has limited output power due to the increased winding loss especially on the secondary side and the high rectifier loss. This issue can be solved by using the

matrix transformer concept [10] [11] [12] where Multiple transformers can be interconnected and the total power is shared between the elemental transformers to reduce secondary winding loss and keep good current sharing between output Synchronous Rectifiers SRs. Here we are exploring the use of the matrix transformer in a 3 kW 400 V to 54 V LLC converter. For this application, the output current ranges from 50 A to 75 A which is a high current to be handled by a single transformer or rectifier. At Least 4 Parallel SR devices must be used but current sharing between SRs cannot be guaranteed. SR Loss is dominated by gate loss and conduction loss. 80V GaN offers low $R_{ds(on)}$ with much less gate charge compared to the Si MOSFET counterpart. Therefore, 80V GaN from EPC is used here. By using a matrix transformer with a series primary winding and parallel secondary winding, the secondary windings will have the same current and this forces symmetry in the SR devices. Increasing the number of elemental transformers per matrix transformer helps to reduce the conduction loss of the secondary winding and SR but also increases the core loss since all elemental transformers are subject to the same volt-sec. Hence, there is a trade-off between the conduction loss and core loss when it comes to designing a matrix transformer. Here we list three possible matrix transformers with different numbers of outputs sets as shown in Fig. 3. All three candidates have single turn secondary winding to reduce conduction loss with high output current.

To get to a conclusion on which transformer topology is better, accurate quantification of core loss is required. Steinmetz model is a popular empirical method to quantify core loss for ferrite material by curve fitting the core loss, which combines Hysteresis, residual, and eddy loss, from a certain ferrite sample to Steinmetz coefficients. Hence, Steinmetz coefficients can predict core loss for a certain ferrite sample within a specific frequency range. However, Steinmetz model fidelity is not guaranteed for a different ferrite sample from the same material with a different core size or shape for the following reasons (dimensional effects):

A. Non-uniform flux distribution:

Steinmetz coefficients are derived from curve fitting of core loss measured of specific ferrite sample. Usually, the ferrite sample is a toroid core with small dimensions (small A_e) to guarantee uniform flux distribution. However, in reality, the

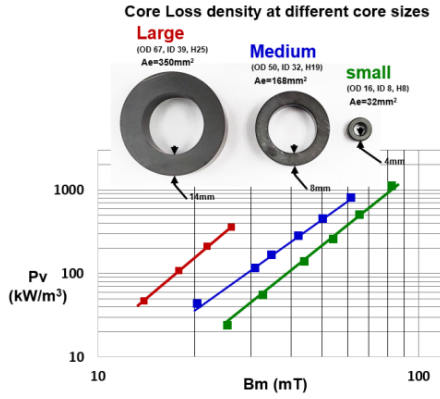


Fig. 4. evaluation of core loss density for different core sizes of the same material (DMR51W)

core size will depend on the application, and also core shape is different. In PCB transformer, UI core or EI core is more convenient to use rather than toroid shape core. Therefore, the flux distribution with the core is non-uniform, and the core loss density is as well non-uniform within the core.

B. Dimensional resonance:

any ferrite material has finite permittivity. This means there's an induced electric field inside the ferrite material. The interaction between electric and magnetic fields tends to redistribute the magnetic field inside the ferrite core following a decaying sinusoidal distribution. If the core thickness is higher than half the wavelength, the magnetic field starts to flow out-of-phase from the excitation current causing drop in core permeability and increased core loss.

C. Skin effect and eddy loss:

any ferrite material has finite conductivity. The source flux induces ac currents flows inside the core. these ac currents are responsible for the eddy loss. These ac currents will also induce eddy fluxes causing the resultant flux to crowd on the surface of the core. The core thickness needs to be smaller than the ferrite skin depth to avoid skin effect.

Fig. 4 shows a case study on 3 different sample sizes from the same material: showing increased core loss density (P_v) with core size. Dimensional resonance and skin effect are highly dependent on the material complex permittivity ($\epsilon = \epsilon' - j\epsilon''$) and permeability ($\mu = \mu' - j\mu''$). These characteristics are frequency-dependent and need to be evaluated at the frequency of interest to get a more accurate estimation of core loss.

The ferrite core material ML91 (from Hitachi) is selected for two reasons: (1) ML91 has the least core loss or least hysteresis loss when measured on a small toroid sample as mentioned in [13]. (2) ML91 has the least AC conductivity value at 1MHz when measured and compared to other materials

TABLE I. COMPARISON OF AC CONDUCTIVITY FOR FERRITE MATERIALS AT 1 MHz

Material	ML91	P63	DMR51W
AC conductivity σ (S/m)	0.5	1	2.5

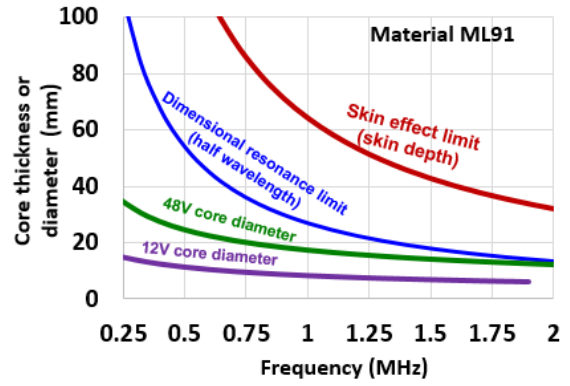


Fig. 5. limits of dimensional resonance and skin effect compared to typical core diameter of 48V transformer and 12V transformer at 50mT

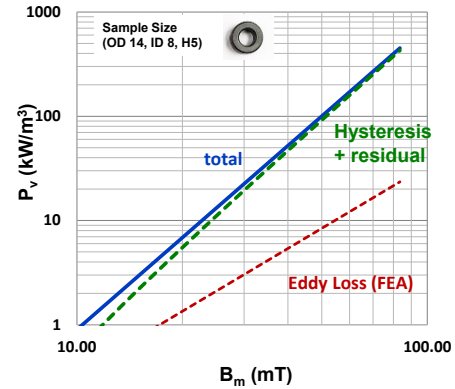


Fig. 6. Breakdown of measured ferrite loss for small toroid sample at 1MHz for ML91

that are optimized for the same frequency as P63 from ACME and DMR51W from DMEGC as shown in table I. The AC conductivity is measured using the parallel plate method. Therefore, ML91 is predicted to have the best performance for the 48V core as it has the least hysteresis loss and the least ac conductivity or eddy loss. The complex permittivity ϵ is measured using the parallel plate method and complex permeability μ is measured using the toroid inductor method [8]. The skin depth and wavelength of the material are evaluated, and the core diameter for typical transformer core for the 48V with 50mT flux density lies below these limits as shown in Fig. 5. Nevertheless, even if the core is designed within the limits, the core still can exhibit higher core loss density compared to the Steinmetz model derived from measured core loss density measured on a small ferrite toroid sample.

According to [9], the core's eddy loss can be calculated for cylindrical ferrite sample as:

$$P_{v_eddy} = \frac{\pi}{4} \sigma A_e f^2 B_m^2 \quad (1)$$

Where σ is material conductivity ($\sigma = \sigma_{dc} + \omega \epsilon'' \epsilon_0$), A_e sample cross-section area. It's clear here that the eddy loss component of ferrite is dependent on the cross-section area of the core.

The methodology to quantify transformer core loss is to, first, acquire core loss measurement ($P_v = P_{v_{Hysteresis}} + P_{v_{residual}} + P_{v_{eddy}}$) of the ferrite sample on a small toroid core as seen in Fig. 6. Since the A_e of the small toroid sample is small, the measured eddy loss is also small, and the measured loss does not reflect the loss in the actual converter core. To get a better estimation of the core loss, the eddy loss is quantified and subtracted from the measured core loss density to single out the $P_{v_{Hysteresis}} + P_{v_{residual}}$. Then, New Steinmetz coefficients are calculated to curve fit $P_{v_{Hysteresis}} + P_{v_{residual}} = k' f^{\alpha'} B^{\beta'}$. The new core loss model will be then calculated as:

$$P_v = k' f^{\alpha'} B^{\beta'} + \frac{\pi}{4} \sigma A_e f^2 B_m^2 \quad (2)$$

Eq. (2) can provide more accurate core loss density and account for eddy loss for different core sizes for sinusoidal excitation. For rectangular excitation as in the case of LLC converter, [9] and [13] agree that the core loss should be multiplied by the factor of $\frac{8}{\pi^2}$. The 2nd term in (2) can be estimated based on the mathematical model for round or square core shape, or from FEA simulation for better accuracy. To account for the non-uniform flux, FEA simulation is used here to account for eddy loss for different core shapes.

III. DESIGN OF MATRIX TRANSFORMER

The three matrix transformer candidates are evaluated based on the EI core shape for each elemental transformer as shown in Fig. 7. Each elemental transformer is composed of 6-layer PCB where the top and bottom layers are two parallel 1-turn secondary windings. Layers 2, and 5 are for EMI shielding layers, and layers 3, and 4 are for primary layers. The EI core has a round center leg. The A_e of the side legs and top and bottom plates are half of the center leg's A_e .

The transformer dimensions are optimized like the process in [12]. The loss for each elemental transformer is evaluated at different combinations of winding width “c” and core radius “r”.

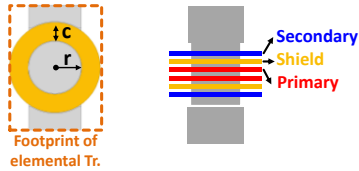


Fig.7. Elemental transformer on EI core and 6-layer PCB

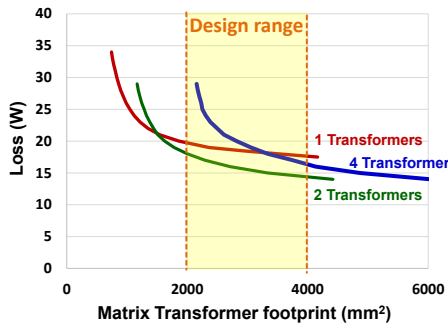


Fig.8. magnetic loss evaluation of different matrix transformer candidates

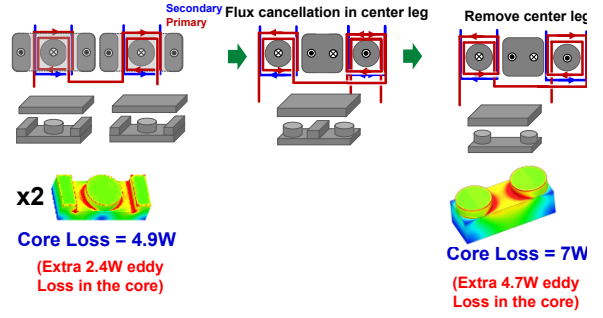


Fig.9. Integration of two EI core transformers and corresponding core loss at 57 mT, $A_e=225 \text{ mm}^2$

By sweeping c and r, the footprint is swept as well. At each transformer footprint, there is a certain “c” and “r” that corresponds to the least magnetic loss at that footprint. The process is repeated at different footprints and the losses are obtained at a wide range of footprints. The two transformers approach showed a lower total loss. Fig. 8. Shows the 2-transformers approach represent the best trade-off between the core loss of matrix transformer and winding loss at the 3 kW power level for the 48 V LLC converter.

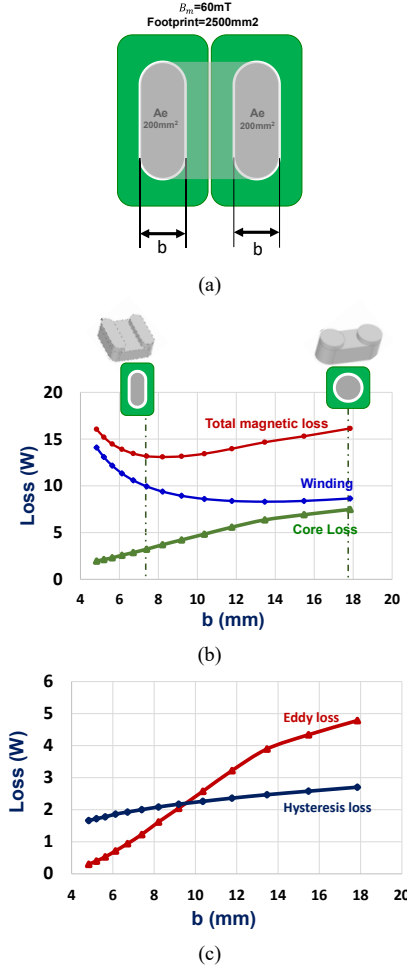


Fig.10. optimization of core thickness to reduce eddy loss (a) core shape (b) transformer loss breakdown, (c) core loss breakdown

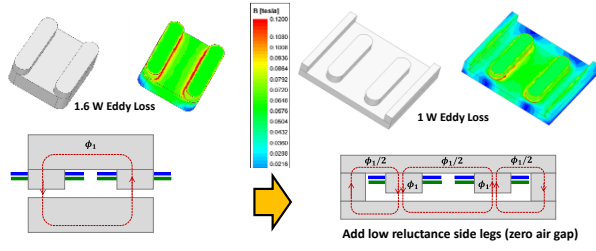


Fig.11. UI core with two side legs

IV. MAGNETIC INTEGRATION OF MATRIX TRANSFORMER

The previous study in section III is done based on a separate EI core for each elemental transformer. However, integrating the elemental transformers into one core has the benefit of easier assembly, lower component count, and less footprint. The two EI cores of the two elemental transformers are integrated into a single UI core following the concept of flux cancellation [5], as shown in Fig. 11. However, after integration, the UI core exhibits higher core loss because of the increased eddy loss. This is explained through (2) as in UI core, the plate thickness is double the plate thickness of EI core, hence higher eddy loss.

The eddy loss in the ferrite follows the same physical concept as eddy loss in copper windings. In windings, the eddy loss appears due to eddy currents induced in the windings by the leakage flux imposed on the winding. The problem is solved by restricting the eddy currents by reducing the winding cross-section area. This is done by using stranded Litz wire or planar winding. The smaller the strands or thickness of planar winding, the less ratio of ac eddy loss to dc loss. Similarly, in ferrite, the eddy currents are induced due to the source flux in the core. Increasing flux density in the core or increasing cross-section area results in higher eddy loss. The problem is solved here using:

A. Rectangular thin core shape:

The increased eddy loss in the UI core is solved by elongating the UI core, so the cross-section area would have a rectangular shape instead of a cylindrical shape. This method reduces the effective area the eddy currents circulate through. This solution is like using planar winding to reduce eddy loss in transformer windings. The core dimensions are optimized for the selected footprint of 2500mm². The core thickness b is swept at the same A_e , B_{max} , and total transformer footprint as in Fig. 10(a). Fig. 10 (b) shows the reduction in core loss as the thickness of the core area reduces (or “ a ” increases). The reduction in core loss is mainly due to eddy loss reduction as depicted in Fig. 10(c). reducing core thickness reduces eddy loss in the transformer legs and top and bottom plates as well. However, reducing the core thickness increases winding loss as well as the winding mean length increases. An optimized core thickness is selected to get the least total transformer loss.

B. Add side legs for better flux distribution:

As mentioned earlier, EI core has inherently less eddy loss than UI core because of the reduced plate area as shown in Fig. 9. The benefit of the EI core can still be utilized in the integrated two transformers by adding two side legs to the UI core as shown in Fig. 11. Adding extra unwinded legs has been introduced

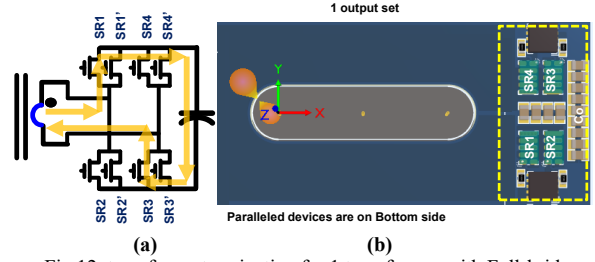


Fig.12. transformer termination for 1 transformer with Full-bridge rectifier and parallel devices (a) circuit diagram, (b) PCB

before in [10], [15] to carry residual flux of the integrated matrix transformer. Here, the side legs are designed with low reluctance (zero airgaps) to force the transformer flux to be shared between the closest side leg and the other transformer. This helps to reduce the flux going through the plate and the plate thickness is further reduced to reduce eddy loss and achieve a thinner core profile.

V. TERMINATION OF FULL-BRIDGE RECTIFIER WITH PARALLEL DEVICES

Another aspect to look into when designing high frequency transformer is the termination loss. The termination means the connection from the transformer winding to the SR devices as shown in Fig 12 (a). This connection lack interleaving and has high AC loss at high frequency and high currents. Also, since we reduced the number of transformers from 4 to 2 transformers, we have to parallel SR devices. In each transformer, we have two full-bridge rectifiers in parallel. This also adds more challenge as the device layout becomes even harder and more termination loss appears.

To keep a symmetric layout between the parallel SRs, we layout one FB on the top side of the PCB and the parallel FB on the bottom side of the PCB as shown in Fig. 12(b). Then, The

TABLE II. SIMULATED AC LOSS AND INDUCTANCE OF FULL-BRIDGE TERMINATION

Termination method	lateral	vertical
AC Loss at 1MHz (3D FEA)	8 W	3.2 W
loop inductance - DC (Q3D)	5.4 nH	4.2 nH
loop inductance - AC (Q3D)	3 nH	1.5 nH

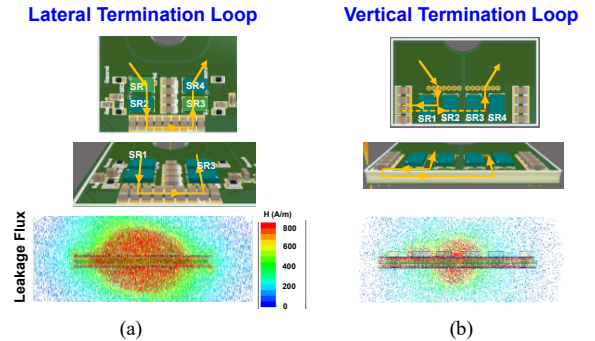


Fig.13. Full-bridge termination and corresponding leakage flux (a) lateral loop, (b) vertical loop

devices of the FB rectifier are laid out as close as possible to each other to reduce the connection loss, the termination loop moves laterally on the top PCB layers, hence the name “lateral termination loop”. The size of the loop is limited by the device’s footprint.

A vertical termination loop is proposed here to reduce the termination loss. The termination loop is folded into two layers so that the path of the current going to the devices overlaps with the current coming out from the devices. Fig. 13(b) shows the vertical termination loop where the SR devices are laid out on the same horizontal lines, and the termination loop goes to SR1 and Co on the top layer and comes back on the second layer (layer 2) to go to SR3. a similar concept has been introduced before in [16] to reduce switching loop inductance for buck converter and applied here to reduce the full-bridge termination loss. This termination strategy provides partial interleaving for the termination current, and we get a smaller termination loop and smaller ac loss.

VI. EXPERIMENTAL RESULTS

The converter is implemented on 6-Layer PCB, 2 oz. copper. 35mΩ-600V GaN devices are used for the primary full-bridge, and 4mΩ, 80V GaN devices are used for the secondary full-bridge rectifiers. Fig 14. shows the implemented PCB including primary and secondary devices and gate drivers and magnetics. The resonant inductor is designed on UI core with side legs as well and the same thickness as the transformer. The inductor and transformer cores are integrated into one core as shown in Fig. 14.. The converter has variable output voltage from 40 V:60 V to charge the bus battery. Fig. 15 shows the converter tested efficiency compared to the first design with four transformers and a round core shape [12]. The proposed core design offers higher light-load efficiency thanks to the reduced core loss at 48 V output.

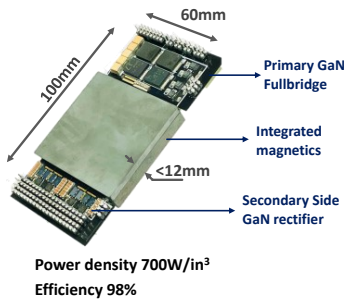


Fig.14. Implemented 1MHz 3kW LLC.

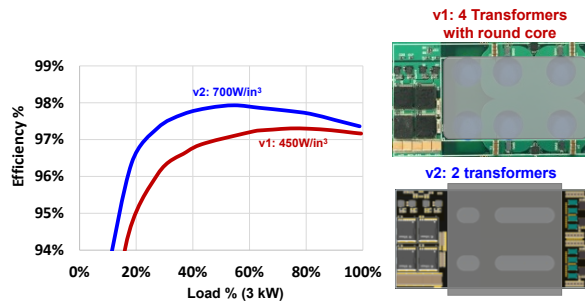


Fig.15. Tested efficiency at Vo = 50 V

VII. CONCLUSIONS AND FUTURE WORK

This paper discussed the impact and importance of dimensional effects on core loss for high power transformer running at high frequency. Different candidates of matrix transformers are evaluated. A matrix of 2-transformers is implemented on UI core with elongated core shape and side legs to reduce core’s eddy loss. The vertical termination loop is proposed to reduce the termination loss of high frequency full-bridge rectifier with parallel devices by providing partial interleaving for the termination current. A prototype of the proposed converter is demonstrated with a low-profile of 12 mm, power density of 700 W/in³, and 98% efficiency.

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